

M31 — Multi-patient wizard depth + per-encounter QR codes

Phase: Phase 7 follow-on (post-M30) **Status:** **DONE** **Blocked by:** M5 (room wizard), M6 (mode toggle), M9 (chat join code), M22 (Per-Patient Console scaffold), M30 (per-encounter parity) **Blocks:** none (closes the multi-patient parity loop with the single-patient ops view) **Estimated effort:** 1 day

1. Purpose

In Room-of-N mode, the multi-patient wizard previously asked the instructor for a **single** set of choices that applied to every bed: one persona list, one curriculum module set, one program/week. It also dropped the instructor at the dashboard with no per-encounter QR codes — every bed shared the same "join the room" QR.

M31 closes the parity loop with the single-patient ops view. The wizard now lets the instructor open three drawers per row (Scenario / Characters / Curriculum), each authoring per-bed:

- **Scenario** — already present from the M30-era bugfix (per-row scenario name + EHR + primary patient persona + chart-mode + label).

- **Characters** — multi-select of the 24 canonical personas, so

Bed 1 can be `patient + MD + charge RN` while Bed 2 is `patient + parent`. The primary patient persona is always included automatically.

- **Curriculum** — per-bed program (ADN-RN, BSN-RN, LPN),

week number, and module set (M01...), so Bed 1 can run a Week-4 sepsis encounter while Bed 2 runs a Week-6 peds encounter.

After the room launches, each Per-Patient Console now renders a **three-cell QR card** (Chat / EHR / Device stations) keyed to that bed's `join_code`, identical in pattern to the v6 ops view but scoped to one encounter. Stations scan once; the join code authenticates them for **this** encounter only — no cross-bed contamination.

This is what closes the user's stated requirement: *"each character in the multi patient can have the same ability assign and operate and instructor has the same level of control as they do on the single with each character on the encounter page."*

2. Structure

Files touched:

- `portal/templates/control.html` — wizard step 4r (room mode) now

exposes `modulesForRoom + programsForRoom` on `window.MEDSIM2`, plus `roleGroup + safetyClass` on `personasForRoom`.

- `portal/static/control.js` — `renderRoomEncounterRows` extended

to render the per-row Scenario / Characters / Curriculum tab strip and drawers; `submitRoom` reads the per-row state and ships it inside each encounter object.

- `portal/static/control.css` — `.encounter-row-tabs`,

`.row-tab[-count]`, `.characters-drawer`, `.curriculum-drawer`, `.row-persona-grid`, `.row-module-grid`, `.curriculum-row`.

- `portal/templates/encounter_console.html` — adds card-network with three `` cells and a copyable URL under each.
- `portal/static/encounter_console.css` — `.qr-grid`, `.qr-cell`, `.qr-label`, `.qr-img`, `.qr-url`; mobile breakpoint collapses to one column.
- `portal/server.py` — `portal_room_encounter_console` adds `"base_url"` to the template context (re-uses the existing `_base_url_for_qr` helper).

No new modules. No schema migration. No new public dataclass fields — `Encounter.selected_personas`, `selected_modules`, `program_id`, and `week` already existed; the wizard simply now *populates them per row* instead of broadcasting a single set.

3. Uses

3.1 Instructor flow

1. Wizard Step 0 → choose **Room of N**. 2. Step 4r — for each row: - **Scenario tab** (default open) — pick `scenario_name`, `patient_persona_id`, `ehr_id`, `chart_mode`, `label`. - **Characters tab** (collapsed) — opens a checkbox grid of the 24 personas (`label` + `role group` + `safety class`). The primary patient persona is always counted, even when its checkbox is hidden in the row. - **Curriculum tab** (collapsed) — `program` `<select>` (defaults to wizard-wide program), `week` `<input type=number>` (defaults to wizard-wide week), and a module-checkbox grid (initialized from wizard-wide modules). 3. Tab badges show live counts (`Characters · 3`, `Curriculum · 2`). 4. Submit → each encounter object now ships its own `personas`, `modules`, `program_id`, `week`.

3.2 Console QR card

After room launch, the instructor opens `/portal/room/encounter/{id}` and sees the **■ Connection · QR codes** card alongside telemetry, ECG, devices, transcript, voice, etc. Each QR cell embeds an `/api/qr.svg?data=...` SVG of the *bed-scoped* join URL:

- `{base_url}/join?code={join_code}` — chat station
- `{base_url}/ehr/join?code={join_code}` — EHR station
- `{base_url}/device/join?code={join_code}` — device station

Mobile devices scan once; the join code is the same one that gates all `/api/encounter/{id}/...` calls for this bed.

4. Functions (exported API surface)

No new public functions. M31 is a wiring / template / JS-glue module — it leans on these already-shipped surfaces:

Symbol	Where	Purpose
<code>POST /api/room/start</code>	<code>portal/server.py</code>	Already accepts <code>personas</code> , <code>modules</code> , <code>program_id</code> , <code>week</code> per encounter; wizard now sends them.
<code>Encounter.selected_personas</code>	<code>portal/control_session.py</code>	List of persona IDs (already existed; populated per row).
<code>Encounter.selected_modules</code>	<code>portal/control_session.py</code>	List of module IDs (already existed; populated per row).

Symbol	Where	Purpose
<code>Encounter.program_id / .week</code>	<code>portal/control_session.py</code>	Curriculum context (already existed; populated per row).
<code>GET /api/qr.svg?data=...</code>	<code>portal/server.py</code> (M5-era)	Returns an SVG QR encoding the URL; called by each <code></code> in the console QR card.
<code>_base_url_for_qr(request)</code>	<code>portal/server.py</code>	Builds the <code>scheme://host[:port]</code> prefix that the template embeds in each station URL.

5. Limitations

- **Wizard does not validate** that a row's primary patient persona

is in the row's character multi-select. The submit handler *unions* the primary persona into the list, so the instructor cannot accidentally launch a bed without its patient.

- **The Curriculum drawer's week field is freeform.** Bed-level

weeks outside the program's published range are accepted silently — they only act as labels in the cohort debrief facet header.

- **QR cells are static SVG.** Refreshing the encounter console

re-issues the same join URLs — the join code does not rotate per scan, matching the v6 ops-view threat model (single-instructor, LAN-only, classroom session). A future module could rotate codes on a per-station first-scan if cross-room contamination becomes a real risk.

- **Module list shown per row is the global library.** There is no

filter "modules a Week-6 BSN student would actually see" — the instructor is trusted to pick coherently. Future M32 could intersect the row's program/week with the module catalog.

- ****QR images do not include a fallback short code or numeric**

pairing PIN.** If the camera can't read the QR, the instructor must type the URL from the `<code class="qr-url">` under each cell.

6. Test status

Test file	Asserts	Status	Last run
<code>tests/v7/test_wizard_depth_and_qr.py::test_encounter_console_renderers_qr_card</code>	card-network + 3 station labels + <code>/join?code=...</code> , <code>/ehr/join?code=...</code> , <code>/device/join?code=...</code> patterns + join code echo in HTML	PASS	2026-05-27
<code>tests/v7/test_wizard_depth_and_qr.py::test_wizard_exposes_modules_and_programs_to_js</code>	<code>modulesForRoom</code> + <code>programsForRoom</code> + <code>roleGroup</code> appear in <code>/portal/control</code> HTML	PASS	2026-05-27

Test file	Asserts	Status	Last run
tests/v7/test_wizard_depth_and_qr.py::test_room_start_carries_per_row_personas_modules_program_week	Two encounters posted with distinct personas/modules/program/week → room.encounters carries per-bed values + no bleed	PASS	2026-05-27
tests/v7/test_wizard_depth_and_qr.py::test_qr_svg_route_responds	/api/qr.svg?data=hello+world returns 200 + image/svg + <svg> body	PASS	2026-05-27
Full v7 suite	200 passed, 1 skipped (M20 Playwright skip — unchanged)	PASS	2026-05-27

7. Change list

Date	Author	Change	Files
2026-05-27	claude-code	QR endpoint param fix (text= → data=) in template + test to match existing /api/qr.svg route signature	portal/templates/encounter_console.html, tests/v7/test_wizard_depth_and_qr.py
2026-05-27	claude-code	Per-row Characters + Curriculum drawers in wizard step 4r	portal/templates/control.html, portal/static/control.js, portal/static/control.css
2026-05-27	claude-code	QR-codes card on Per-Patient Console (chat / EHR / device)	portal/templates/encounter_console.html, portal/static/encounter_console.css, portal/server.py (template context)
2026-05-27	claude-code	4 acceptance tests for both halves of M31	tests/v7/test_wizard_depth_and_qr.py

8. Open questions / known issues

- ****The Characters drawer multi-select uses the full 24-persona**

library.** When the row's primary persona is e.g. a pediatric patient, the drawer still shows e.g. the CCU-RN — there is no contextual filter. The wizard's Step-4 (single-patient) filters apply only to the wizard-wide list, not the per-row drawer. This is intentional for v7.0 (the instructor knows the cast they want) but a future M32 could mirror Step-4's `roleGroup` + `safetyClass` filters inside the row drawer.

- **The QR card refresh is page-load only.** If a join code is

rotated mid-session (future feature), the instructor must hard-refresh the console to see the new QR. The existing 2-s poll in `encounter_console.js` does not touch the QR card.

- **Mobile collapse rule** uses `@media (max-width: 700px)` —

matches the v6 ops view; verify if the target classroom tablet is a 10" Android (likely fine) vs. a 7" model (cells stack early, which is acceptable).

- **No telemetry on the QR card.** Per M30 design we kept the

cards independent — telemetry has its own card, devices have theirs, network has its own. If we ever consolidate, the QR card should remain visible (instructor's primary "how do students get in" landmark).

Render this guide to PDF with `python docs/module_guides/render_pdfs.py` (uses the project's installed `pandoc / wkhtmltopdf` pipeline). The Markdown is the source of truth — regenerate the PDF on each material change.